Fig. 1. ThermalSight Clinical Thermobiomechanics & DFU Screening Workflow

Stage 1: Radiometric Ingestion, Inpainting Artifact Removal & Plantar Anatomical Segmentation (Phases 1–3)

Phase 1: Radiometric FLIR Ingestion & Thermal Scale Calibration
- Direct extraction of calibrated °C via flyr FLIR unpack
- True physiological temperature matrix [20.0°C–42.0°C]
- Adaptive luminance-to-Celsius fallback for grayscale
- Automated elimination of >100°C measurement anomalies

Phase 2A: OSD Accessory Inpainting & Sub-Degree Float32 Precision
- Bounding-box masking of camera OSD & crosshairs
- Telea inpainting maintaining strict float32 decimals
- Hotspot protection avoiding dark-hole tissue deletion
- Full preservation of subtle sub-degree plantar gradients

Phase 2B: Plantar Morphology & Background Edge Isolation
- Automated 31.0°C background cutoff segmentation
- Morphological kernel (7x7) smoothing plantar boundaries
- Generation of strict binary foot mask (mask_dense)
- Exclusion of cold background leakage into toe borders

Phase 3A: Standardized Clinical ROI Registration & Scaling
- Multi-point landmark mapping: Hallux (T1), Metatarsals (M1–M3), Heel
- Multi-polygon & circular ROI delineation with 36-point boundary vertices
- Physical distance calibration via spatial scale factor (pxPerCm)
- Session persistence via annotations_session.json for longitudinal tracking

Phase 3B: High-Resolution Spatial Thermal Topography
- 1:1 Full-grid continuous temperature matrix sampling across plantar skin
- 2D Gaussian kernel smoothing & 11-level isothermal contour elevation
- Turbo colormap normalization establishing standardized thermal contrast
- Sub-pixel coordinate registration eliminating 0.5-cell offset artifacts

Stage 2: Quantitative Triad Evaluation (PPP, PPG, PGA) & Clinical Diagnostic Decision Support (Phases 4–5)

Phase 4A: PPP & PPG Biomechanical Magnitude Extraction
- PPP (Plantar Peak Perfusion): Absolute localized T_peak, T_mean, T_min, std
- 9x9 local patch sampling strictly filtered within foot mask (mask_dense)
- PPG (Plantar Thermal Gradient): Spatial thermal slope $||\nabla T|| = \sqrt{G_x^2 + G_y^2}$
- 2D Sobel spatial differentiation capturing acute inflammatory boundaries

Phase 4B: PGA Directional Vector & Coherence Analysis
- PGA (Gradient Angle): Vector orientation $\theta = \arctan2(-G_y, G_x) \bmod 360°$
- Corrected compass orientation: 90° = North (Toes/Distal), 270° = South (Heel)
- Circular vector coherence metric (0–100%) quantifying heat flow uniformity
- Separation between uniform vascular dissipation vs. multidirectional anomaly

Phase 5A: Multi-Modal Topographic Visualization Engine
- 2D Vector Quiver Field mapping localized heat dissemination trajectories
- 3D Thermal Elevation Surface meshes pinpointing focal hyperemic summits
- 8-Point Star Radar charts evaluating directional steepness (N, NE, E...)
- Publication-ready vector outputs (Fig 1 Panel A/B) at 300 DPI print quality

Phase 5B: Automated Audit Trail & Statistical Synthesis
- Cross-subject cohort aggregation exporting detailed long & wide CSV summaries
- Multi-sheet Excel workbook with live formulas (=AVERAGE, =MAX, =MIN)
- Transparent methodology guide sheet enabling reproducible scientific peer review
- Headless CLI non-blocking batch execution for automated research pipelines

Bilateral Asymmetry & Inflammation Triage
- Contralateral temperature difference screening ($\Delta T > 2.2°C$ threshold)
- Separation of diffuse physiological warmth from acute osteomyelitis borders
- Identification of sub-clinical tissue stress prior to visible dermal ulceration

Phase 5C: Clinical Decision Support & DFU Prevention Policy
- Early risk stratification for proactive Diabetic Foot Ulcer (DFU) prevention
- Quantitative objective biomechanical evidence guiding targeted footwear offloading
- Longitudinal wound healing monitoring and tele-thermology clinical decision support